# Dax L. Feliz, PhD

Curriculum Vitae

Flatiron Institute
Center for Computational Astrophysics
New York, NY 10010

dfeliz@flatironinstitute.org
daxfeliz.github.io
0000-0002-2457-7889


Last updated: June 1, 2026

## Research Interests

Developing an end-to-end observational program to forecast atmosphere retention in nearby M-dwarf planetary systems, combining: (1) completeness-controlled transit surveys of M-dwarfs in TESS FFIs (NEMESIS); (2) multi-epoch flare frequency distributions and stellar activity evolution from TESS, K2, and ground-based photometry; and (3) Galactic-context characterization via Gaia kinematics and SDSS spectroscopic metallicities. Long-term goal: a ranked, observation-ready target list for atmospheric characterization with JWST, Roman, and PLATO. This program is enabled by careful photometric performance characterization, including end-to-end systematics modeling, transit injection-recovery, and completeness-controlled detection-efficiency mapping at the pixel-to-light-curve level.

## Education

- **PhD Astrophysics**, Vanderbilt University, May 2022. Advisor: Dr. Keivan Stassun. Dissertation Link
- **M.A. Physics**, Fisk University, May 2018. Advisors: Drs. Keivan Stassun & Karen Collins.
- **B.S. Astrophysics**, University of Massachusetts Amherst, May 2013. Advisor: Dr. Min Yun.

## Academic Appointments

**Flatiron Research Fellow, Center for Computational Astrophysics (CCA), Sep 2024 – Present**
*Postdoctoral Advisor: Dr. David Hogg*,

- Developing an independent research program measuring exoplanet occurrence rates and flare frequency distributions around nearby M-dwarfs using NASA TESS and ground-based photometry; extending the NEMESIS pipeline for FFI-based transit surveys; initiated and co-supervised one CUNY Bridge graduate thesis project and one undergraduate research project.

**Postdoctoral Researcher, American Museum of Natural History (AMNH), Sep 2022 – Sep 2024**
*Postdoctoral Advisor: Dr. Ruth Angus*,

- Led independent transit and stellar activity analyses of nearby M-dwarfs using TESS Full Frame Images photometry; initiated and co-supervised one CUNY Bridge graduate thesis project, two undergraduate internship projects and six high school internship projects.

**Postdoctoral Researcher, Vanderbilt University, May 2022 – Aug 2022**
*Postdoctoral Advisor: Dr. Keivan Stassun*,

**Visiting Graduate Student Fellow, George Mason University, 2021 – 2022**

*Advisors: Drs. Peter Plavchan & Keivan Stassun*

## Selected Research Contributions / Research Program 🌎

**Atmosphere Retention Forecasting for Nearby M-dwarf Planets, CCA 2024 – Present**

- Developing an end-to-end observational program to forecast atmosphere retention in nearby M-dwarf planetary systems, combining: (1) completeness-controlled transit surveys of M-dwarfs in TESS FFIs (NEMESIS); (2) multi-epoch flare frequency distributions and stellar activity evolution from TESS, K2, and ground-based photometry; and (3) Galactic-context characterization via Gaia kinematics and SDSS spectroscopic metallicities. Long-term goal: a ranked, observation-ready target list for atmospheric characterization with NASA JWST, NASA Roman, and ESA PLATO instruments.

**Transit Survey of M-Dwarf Stars in NASA TESS Data, VU/AMNH/CCA 2018 – Present**

- Architect and lead developer of NEMESIS, an end-to-end pipeline that extracts TESS Full Frame Image light curves at the pixel level, models and removes instrumental systematics, and runs BLS-based periodic transit searches over nearby M-dwarf samples. Performance characterization includes injection-recovery testing, per-target detection efficiency maps, and completeness-corrected occurrence-rate estimation. Pipeline extensions include multi-epoch flare detection, flare frequency distribution inference, and Galactic-context cross-matching with Gaia and SDSS.

**ESA PLATO Flare Removal Working Group, European Space Agency, 2020 – Present**

- Member of the ESA PLATO Flare Removal Working Group. Developing and benchmarking algorithms to identify and remove stellar flare events from PLATO photometric time series, with attention to preserving transit signal integrity and minimizing biases in downstream science products. Co-author on the PLATO mission paper (Rauer et al. 2025).

**Kilodegree Extremely Little Telescope (KELT), VU, TN 2017 – 2022**

- Operated KELT data reduction pipeline for detection of transiting exoplanets orbiting bright stars with a ground-based telescope network (Vanderbilt University / The Ohio State University collaboration). Also contributed to ground-based transit follow-up vetting within KELT-Follow Up Network (KELT-FUN), an international observer network whose membership and operational protocols directly seeded the TESS Follow-up Observing Program (TFOP) working group; reviewed reduced photometric data, assessed light curve quality, and contributed to disposition updates for transiting planet candidates.

**Multi-Year Transit Survey of Proxima Centauri, Fisk University & VU, TN 2016 – 2019**

- Master's Thesis: Produced and detrended light curves from Proxima Centauri observations; applied periodic transit search algorithms to search for transit events of the Earth-sized planet Proxima b.

## Fellowships 🏆

- NSF AGEP Fellowship, Spring 2021 – 2022. $57,272. *(competitive national award)*
- Fisk-Vanderbilt Bridge Fellowship, Fall 2018 – 2021. $39,000 annual stipend.

## Successful Proposals / Grants 💰

- NASA Transiting Exoplanet Survey Satellite (TESS) Guest Investigator Cycle 9, *Long-Term Evolution of M-dwarf Flare Activity with K2, TESS, and Ground-Based Photometry* — PI: D. L. Feliz. Submitted

March 2026. *(NASA ROSES; under review)*
- NSF Graduate Opportunities at Fisk in Astronomy and Astrophysics Research (GO-FAAR) grant, 2016 – 2018. NSF Grant #1358862. $20,000 annual stipend.
- McMinn Summer Research Award, 2018. $5,000.
- New York Space Consortium Research Grant, Summer 2012. $5,000.

## Honors & Recognition 🏆

- Heroes Classroom Namesake: The Feliz Room, East Harlem Scholars Academy, New York, NY, 2013.

## Professional Service 🤝

- **Member**, ESA PLATO Flare Removal Working Group, 2020 – Present.
- **Instructor**, CUNY Master's in Astrophysics Bridge Programming Bootcamp, 2022 – Present. Curriculum developer and lead instructor for annual two-week graduate onboarding bootcamp; ~10–15 students per cohort.
- **TOI-Vetter** Served as voluntary vetter for the TESS Objects of Interest (TOI) working group and the TESS Single Transit Vetting working group; reviewed transiting planet candidates, assessed false-positive scenarios, and updated dispositions that contributed to ExoFOP-TESS, a web-based service developed and operated by the NASA Exoplanet Science Institute (NExScI).

## Invited Colloquia & Seminars (4 invited / 2 contributed)

- *'From Exoplanet Detections to Atmosphere Forecasts Around Nearby M-Dwarfs'*, New York City College of Technology Physics Departmental Seminar, Feb 2026.
- *"NEMESIS I: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs I"* — MIT TESS Science Talk (Mar 2021); Rubin LSST TVS Seminar (Jun 2021); Center for Exoplanets & Habitable Worlds Seminar, Penn State (Sep 2021).
- *"How to deflect an incoming asteroid"*, University of Massachusetts Amherst Astronomy Colloquium, May 2013.

## Invited Conference & Workshop Talks (7 invited) 🎤

- *"NEMESIS II: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs"* — Emerging Researchers in Exoplanets (ERES) X, Princeton University (Jun 2025); "From Trends to Transits" Workshop, University of New Mexico (Aug 2025).
- *'M-dwarfs are cool!'*, AstroFest at the Flatiron CCA (Sep 2024 & Sep 2025).
- *"Twinkle, Twinkle Little Star"*, Twinkle and the Next Generation of Exoplanet Scientists Conference, Sep 2021.
- *"Using BLS modeling on photometry of Proxima Centauri"*, KELT Workshop, Lehigh University, Jun 2017.
- *"Light Curve Characterization of High Proper Motion M-Dwarf Stars with the Kepler Space Telescope"*, AMNH REU Annual Symposium, Aug 2012.
- *"Using Spectropolarimetry to Study Extrasolar Planets"*, AMNH REU Annual Symposium, Aug 2010.

## Contributed Talks & Posters (5 contributed)

- *"NEMESIS II: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs"* — TESS Science Conference III, MIT (Jul 2024, Zenodo link here); Sagan Summer Workshop, Caltech (Jul 2025, Poster link here).
- *"Can Photometric Flares Detected by Ground and Space Based Telescopes Be Used To Identify Activity Cycles in M-dwarf stars?"*, Cool Stars 22, Jul 2024 (Zenodo link here).
- *"NEMESIS I: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs I"* — Cool Stars 20.5 (Mar 2021, Zenodo link here); TESS Science Conference II (Aug 2021, Zenodo link here)).

## Mentoring & Advising Experience

Research mentoring across high school, undergraduate, and graduate levels with projects in exoplanet detection, stellar activity, and time-series modeling; trainees have presented at conferences and workshops and have advanced to graduate programs and industry positions. **Student totals:** High School: 7; Undergraduate: 5; Graduate: 2.

### Barnard Individual Research Course (PHYS-BC3900), Flatiron CCA

- *Evaluating Planet Vetting Robustness Across TESS Photometric Pipelines*, Spring 2026. Undergraduate Student: Madeline J. Maldonado Gutierrez, Barnard College of Columbia University.

### CUNY Master's in Astrophysics, Bridge Thesis Project

- *Searching for M-dwarf Coronal Mass Ejections with TESS and SDSS-V*, 2024 – Present. Graduate Student: Andrea Bracamonte, CUNY Graduate Center. Co-advised with Dr. Ruth Angus. Project outputs contribute a framework for producing flare-rate priors and SDSS spectroscopic detection efficiency estimates for the NEMESIS atmosphere-retention forecasting program.

### Barnard Summer Research Institute, AMNH

- *Detection and Validation of Transiting Exoplanets Around Nearby M-dwarf Stars*, May – Aug 2024. Undergraduate Student: Madeline J. Maldonado Gutierrez, Barnard College of Columbia University.

### CUNY Master's in Astrophysics, Bridge Thesis Project

- *Using Gaussian Processes to Model Stellar Rotation in Zwicky Transient Facility Photometry*, 2022 – 2024. Graduate Student: Ryan Lebron, CUNY Graduate Center. Co-advised with Dr. Ruth Angus. Project outputs contribute a GP-derived rotation period catalog used as stellar activity priors in the NEMESIS atmosphere-retention forecasting program.

### AMNH Research Experience for Undergraduates (REU) Program

- *Detecting Exoplanets Around Nearby M-dwarf Stars from the Revised TESS Habitable Zone Catalog*, 2023. Undergraduate Student: Maliyah Adams, Arizona State University.

### AMNH Science Research Mentoring Program

- *Detection of Transiting Exoplanets and Eclipsing Binaries Around Nearby M-dwarf Stars*, 2022 – 2024. High School Students: August Fischer, Donovan Bradley, Jashcelyn Canada (2022–2023); Kylor Ghai, Shreeya KC, Thamim Chowdhury (2023–2024).

### Central American-Caribbean Bridge Program

- *Recovery of Known TESS Objects of Interest from TESS Full Frame Images*, 2020 – 2021. Undergraduate Student: Bryan Villarreal Alvarado, University of Costa Rica.

**Summer Research Internships, Vanderbilt University**

- *Transiting Exoplanet Survey Satellite: A Search for New Worlds*, 2019 – 2020. Undergraduate Student: Samantha Bianco, Vanderbilt University.
- *Blind Transit Survey of TESS Data for M Dwarf Systems*, 2019 – 2020. Undergraduate Student: Mary Jimenez, George Mason University.

**School for Science and Math at Vanderbilt**

- *Detecting Exoplanet Transits in TESS Light Curves*, Spring 2019. High School Student: Felix Bean, Hunter's Lane High School.

## Teaching Experience

**CUNY Master's in Astrophysics Bridge Program: Programming Bootcamp Instructor**, Aug 2022, 2023, 2024 & 2025

- Two-week orientation and programming bootcamp for incoming astronomy graduate students of varied backgrounds; ~10–15 students per cohort. Responsibilities include curriculum design and development, instruction in Python, data analysis, and astronomical software tools, and integration of students into the NYC astronomy community. Course materials publicly available at this link here.

**Teaching Assistantships, Vanderbilt University**

- ASTR-101L: Introductory Solar System Laboratory, Fall semesters 2018 – 2020. *(∼10 hrs/week)*
- ASTR-102L: Introductory Stars & Galaxies Laboratory, Spring semesters 2018 – 2020. *(∼10 hrs/week)*

**Laboratory Instruction / Teaching Support**

- Electronics Laboratory, Bronx Community College, NY, Spring 2014 – Fall 2015. *Full Time: (∼40 hrs/week)*

## Software and Open Science Products

**Research Software & Pipelines (Selected)**

- **NEMESIS** (TESS Full Frame Image transit-photometry pipeline) — Developer, 2018 – Present. Built pipeline to extract TESS FFI light curves, remove instrumental systematics, and run periodic transit searches to identify candidates. Pipeline scope extended to include multi-epoch stellar flare detection, flare frequency distribution inference, and Galactic-context characterization via Gaia and SDSS cross-matching. Capabilities include FFI cutout extraction, aperture optimization, cotrending against pixel-level basis vectors, BLS transit search, and injection-recovery based completeness quantification. *Status: built, in active development.*
- **ESA PLATO flare-removal algorithm** (PLATO Flares Working Group) — Developer/Contributor, 2020 – Present. Developing methods to detect and remove stellar flare events from PLATO mission photometric time series. *Status: built, in active development.*
- **Kilodegree Extremely Little Telescope data reduction pipeline** — Operator/Analyst, 2017 – 2022. Operated pipeline for transit detections in bright-star ground-based survey data. *Status: operated/maintained/applied.*

**Data Products / Catalogs (Selected)**

- **NEMESIS I Planet Candidates** — Catalog of vetted Community TESS Objects of Interest from the NEMESIS I blind-transit survey of TESS Full Frame Images. https://filtergraph.com/NEMESIS

### Computational Training Materials

- Programming Bootcamp tutorials for the CUNY Master's Program in Astrophysics. https://daxfeliz.github.io/cunybridgebootcamp/

## Outreach & Public Engagement

- **East Harlem Scholars Academy**, New York, NY — Alumni Day Speaker, 2010 – 2015. Answered questions from elementary school students about astronomy, physics, and research.
- **Central Park East High School**, New York, NY — College Panel Speaker, 2009 – 2013; Robotics Consultant, 2009 – 2011. Spoke to high school students about science and higher education; volunteered with the FIRST Robotics Competition.
- **East Harlem Tutorial Program**, New York, NY, 2008 – 2010. Tutored and mentored high school and middle school students in math and science; served as a FIRST Robotics competition mentor.

## Publication Metrics

Refereed: 32 / 1st Author: 3 / Citations: 1,061 / h-index: 17 (2026-04-17, via ADS).

## 1st-Authored Refereed Publications

1. **Dax L. Feliz**, Peter Plavchan, Samantha N. Bianco, and ... NEMESIS: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs. I. *The Astronomical Journal*, 161(5):247, May 2021
2. **Dax L. Feliz** L., David L. Blank, Karen A. Collins, and ... A Multi-year Search for Transits of Proxima Centauri. II. No Evidence for Transit Events with Periods between 1 and 30 days. *The Astronomical Journal*, 157(6):226, Jun 2019
3. David L. Blank, **Dax L. Feliz****, Karen A. Collins, and ... A Multi-year Search for Transits of Proxima Centauri. I. Light Curves Corresponding to Published Ephemerides. *The Astronomical Journal*, 155(6):228, Jun 2018, ****Corresponding author**

## Selected Collaborative Refereed Publications

1. Peter Plavchan, Thomas Barclay, Jonathan Gagné, ... **Dax Feliz** , and ... A planet within the debris disk around the pre-main-sequence star AU Microscopii. ***Nature***, 582(7813):497–500, June 2020
2. Heike Rauer, Conny Aerts, Cabrera, ..., **D. L. Feliz**, and ... The PLATO mission. *Experimental Astronomy*, 59(3):26, June 2025
3. Karen A. Collins, Kevin I. Collins, Joshua Pepper, ..., **Dax L. Feliz** , and ... The KELT Follow-up Network and Transit False-positive Catalog: Pre-vetted False Positives for TESS. *The Astronomical Journal*, 156(5):234, Nov 2018

## Nth-Authored Refereed Publications

1. Madison G. Scott, Georgina Dransfield, Mathilde Timmermans, ..., **Dax L. Feliz**, and ... Two temperate Earth- and Neptune-sized planets orbiting fully convective M dwarfs. *Monthly Notices of the Royal Astronomical Society*, January 2026

2. N. Heidari, G. Hébrard, E. Martioli, ..., **D. L. Feliz**, and ... Characterization of seven transiting systems, including four warm Jupiters from SOPHIE and TESS. *Astronomy & Astrophysics*, 694:A36, February 2025

3. E. Semenko, O. Kochukhov, Z. Mikulášek, ..., **D. L. Feliz**, and ... HD 34736: an intensely magnetised double-lined spectroscopic binary with rapidly rotating chemically peculiar B-type components. *Monthly Notices of the Royal Astronomical Society*, 535(3):2812–2836, December 2024

4. Rishi R. Paudel, Thomas Barclay, Allison Youngblood, ..., **Dax L. Feliz**, and ... A Multiwavelength Survey of Nearby M Dwarfs: Optical and Near-ultraviolet Flares and Activity with Contemporaneous TESS, Kepler/K2, Swift, and HST Observations. *The Astrophysical Journal*, 971(1):24, August 2024

5. Lionel J. Garcia, Daniel Foreman-Mackey, Catriona A. Murray, ..., **Dax L. Feliz**, and ... nuance: Efficient Detection of Planets Transiting Active Stars. *The Astronomical Journal*, 167(6):284, June 2024

6. Christopher R. Mann, Paul A. Dalba, David Lafrenière, ..., **Dax L. Feliz**, and ... Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. III. Recovery and Confirmation of a Temperate, Mildly Eccentric, Single-transit Jupiter Orbiting TOI-2010. *The Astronomical Journal*, 166(6):239, December 2023

7. Justin M. Wittrock, Peter P. Plavchan, Bryson L. Cale, ..., **Dax L. Feliz**, and ... Validating AU Microscopii d with Transit Timing Variations. *The Astronomical Journal*, 166(6):232, December 2023

8. Ismael Mireles, Diana Dragomir, Hugh P. Osborn, ..., **Dax L. Feliz**, and ... TOI-4600 b and c: Two Long-period Giant Planets Orbiting an Early K Dwarf. *The Astrophysical Journal Letters*, 954(1):L15, September 2023

9. Keivan G. Stassun, Guillermo Torres, Marina Kounkel, ..., **Dax L. Feliz**, and ... An Eclipsing Binary Comprising Two Active Red Stragglers of Identical Mass and Synchronized Rotation: A Post-mass-transfer System or Just Born That Way? *The Astrophysical Journal*, 950(2):99, June 2023

10. Joseph E. Rodriguez, Samuel N. Quinn, Andrew Vanderburg, ..., **Dax L. Feliz**, and ... Another Shipment of Six Short-Period Giant Planets from TESS. *Monthly Notices of the Royal Astronomical Society*, February 2023

11. Mohammed El Mufti, Peter P. Plavchan, Howard Isaacson, ..., **Dax L. Feliz**, and ... TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs. *The Astronomical Journal*, 165(1):10, January 2023

12. Keivan G. Stassun, Guillermo Torres, ..., **Dax L. Feliz**, and ... A Low-mass Pre-main-sequence Eclipsing Binary in Lower Centaurus Crux Discovered with TESS. *The Astrophysical Journal*, 941(2):125, December 2022

13. S. Ulmer-Moll, M. Lendl, S. Gill, ..., **D. L. Feliz**, and ... Two long-period transiting exoplanets on eccentric orbits: NGTS-20 b (TOI-5152 b) and TOI-5153 b. *arXiv e-prints*, page arXiv:2207.03911, July 2022

14. Ian Stotesbury, Billy Edwards, ..., **Dax L. Feliz**, and ... Twinkle: a small satellite spectroscopy mission for the next phase of exoplanet science. In Laura E. Coyle, Shuji Matsuura, and Marshall D. Perrin, editors, *Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave*, volume 12180 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 1218033, August 2022

15. Justin M. Wittrock, Stefan Dreizler, Michael A. Reefe, Brett M. Morris, ..., **D. L. Feliz**, and ... Transit Timing Variations for AU Microscopii b and c. *The Astronomical Journal*, 164(1):27, July 2022

16. Michael A. Reefe, Rafael Luque, Eric Gaidos, ..., **D. L. Feliz**, and ... A Close-in Puffy Neptune with Hidden Friends: The Enigma of TOI 620. *The Astronomical Journal*, 163(6):269, June 2022

17. Jiayin Dong, Chelsea X. Huang, George Zhou, ..., **D. L. Feliz**, and ... NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star. *The Astrophysical Journal Letters*, 926(2):L7, February 2022

18. Christina Hedges, Alex Hughes, George Zhou, ..., **D. L. Feliz**, and ... TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up. *The Astronomical Journal*, 162(2):54, August 2021

19. Keivan G. Stassun, Guillermo Torres, Cole Johnston, ..., **Dax L. Feliz**, and ... Discovery and Characterization of a Rare Magnetic Hybrid $\beta$ Cephei Slowly Pulsating B-type Star in an Eclipsing Binary in the Young Open Cluster NGC 6193. *The Astrophysical Journal*, 910(2):133, April 2021

20. S. A. Rappaport, D. W. Kurtz, G. Handler, ..., **D. L. Feliz**, and ... A tidally tilted sectoral dipole pulsation mode in the eclipsing binary TIC 63328020. *Monthly Notices of the Royal Astronomical Society*, February 2021

21. Joseph E. Rodriguez, Samuel N. Quinn, George Zhou, ..., **Dax L. Feliz**, and ... TESS Delivers Five New Hot Giant Planets Orbiting Bright Stars from the Full-frame Images. *The Astronomical Journal*, 161(4):194, April 2021

22. Joni-Marie C. Cunningham, **Dax L. Feliz**, Don M. Dixon, and ... A KELT-TESS Eclipsing Binary in a Young Triple System Associated with the Local "Stellar String" Theia 301. *The Astrophysical Journal*, 160(4):187, October 2020

23. Romy Rodríguez Martínez, B. Scott Gaudi, Joseph E. Rodriguez, ..., **Dax L. Feliz** , and ... KELT-25 b and KELT-26 b: A Hot Jupiter and a Substellar Companion Transiting Young A Stars Observed by TESS. *The Astronomical Journal*, 160(3):111, September 2020

24. Weicheng Zang, Subo Dong, Andrew Gould, . .., **Dax L. Feliz**, and . .. Spitzer + VLTI-GRAVITY Measure the Lens Mass of a Nearby Microlensing Event. *The Astrophysical Journal*, 897(2):180, July 2020

25. Joseph E. Rodriguez, Jason D. Eastman, George Zhou, ..., **Dax L. Feliz** , and ... KELT-24b: A 5M $_J$ Planet on a 5.6 day Well-aligned Orbit around the Young V = 8.3 F-star HD 93148. *The Astronomical Journal*, 158(5):197, Nov 2019

26. Jonathan Labadie-Bartz, Joseph E. Rodriguez, Keivan G. Stassun, ..., **Dax L. Feliz** , and ... KELT-22Ab: A Massive, Short-Period Hot Jupiter Transiting a Near-solar Twin. *The Astrophysical Journal*, 240(1):13, Jan 2019